

Business Insights & Recommendations

1. Executive Summary

This analysis examines Walmart sales data to understand how demand is distributed across products, categories, stores, and time, with particular attention to patterns that can influence demand and inventory planning. The analysis focuses on demand concentration, store performance, category contribution, demand variability, intermittent sales activity, and changes in demand associated with holidays and events. The purpose is to translate these patterns into practical business insights that can support more focused planning and decision-making.

A major finding is that **demand is highly concentrated among a relatively small group of products**. The top 10% of products account for **51.14% of total unit demand**, while approximately 38% of products account for around 80% of total demand. This indicates that a relatively small portion of the product portfolio contributes a substantial share of overall demand. The analysis also shows that the **FOODS category contributes 68.63% of total unit demand**, making it the largest contributor among the three categories examined.

The analysis further shows that demand volume and demand stability are not always the same. Several store-product combinations were identified as having both **above-average demand and above-average demand variability**. These products can generate substantial demand while also experiencing considerable fluctuations, making them important candidates for closer forecasting and replenishment attention. At the other end of the demand spectrum, some products exhibit highly intermittent sales activity. For example, **FOODS_3_296 recorded sales on only 10.04% of the 1,913 days covered by the dataset**, indicating that its demand pattern differs substantially from products with more consistent sales activity.

Taken together, these findings suggest that **a uniform approach to demand and inventory planning may not be equally appropriate across all products and stores**. Greater attention can be directed toward high-contribution products, high-demand products with substantial variability can be monitored more closely, and intermittent products can be evaluated using planning approaches suited to infrequent demand. Store-level performance differences and observed holiday/event patterns can provide additional context for more localized and event-aware planning. These findings should be viewed as **demand-planning signals rather than evidence of confirmed stockouts or inventory shortages**, because actual inventory and stockout data were not available for this analysis.

2. Key Business Insights

2.1 Demand Is Highly Concentrated Across Products

The product-level analysis revealed a strong concentration of demand within the portfolio. Rather than being distributed evenly across all products, a relatively small group of products accounts for a substantial share of total unit demand. The analysis covered **3,049 products** and showed that the **top 10% of products contribute 51.14% of total demand**. Furthermore, approximately **38% of the products account for around 80% of total demand**.

This finding is important because it identifies a clear **priority structure within the product portfolio**. A relatively small number of products has a disproportionately large influence on overall demand. As a result, forecasting errors, replenishment issues, or inadequate planning for these products could have a greater impact on overall demand fulfillment than similar issues involving lower-volume products.

The analysis therefore provides a practical basis for **demand-based product prioritization**. Instead of allocating the same level of planning attention across the entire product range, management can place greater emphasis on products that contribute the largest share of demand. These products can be given closer monitoring, more frequent review of demand patterns, and greater attention during replenishment and forecasting activities.

Business Implication

The key implication is that **product importance is not evenly distributed across the portfolio**. Demand concentration provides a data-driven basis for identifying products that deserve greater planning attention.

Recommendation

Walmart could use demand contribution as one factor in **product segmentation and planning prioritization**. High-contribution products should receive greater forecasting and replenishment attention, while lower-contribution products can be managed with appropriately scaled planning effort. This would allow resources to be focused where they are likely to have the greatest impact on overall demand.

2.2 FOODS Is the Dominant Demand Category

The category-level analysis shows that demand is strongly concentrated in the **FOODS** category. Across the full analysis period, FOODS generated **45,089,939 units**, representing **68.63% of total unit demand**. HOUSEHOLD contributed **22.04%**, while HOBBIES accounted for **9.32%**.

This distribution shows that FOODS is not simply the largest category—it is the **primary driver of overall demand volume** in the dataset. Consequently, changes in demand within FOODS have a much greater potential impact on overall demand patterns than comparable changes in the smaller categories.

Business Implication

The concentration of demand in FOODS provides a clear basis for **category-level prioritization**. Demand planning, forecasting, and replenishment decisions within this category are particularly important because a large proportion of the overall unit demand is associated with it.

At the same time, the result should not be interpreted as meaning that HOUSEHOLD and HOBBIES are unimportant. Their smaller shares indicate that they contribute less to total unit volume, but they may still contain individual products or store-level patterns that require attention.

Recommendation

Walmart could give **greater planning and monitoring attention to the FOODS category**, particularly its high-demand products. Category-level demand trends can be used alongside product-level concentration and variability analysis to prioritize forecasting and replenishment efforts.

2.3 High Demand Can Come with High Variability

The analysis of store-product combinations revealed an important planning pattern: **products with high average demand are not necessarily stable in their day-to-day demand**. To identify these cases, the analysis first compared each store-product combination with the overall average daily demand of **1.13 units**. Among the 30,490 store-product combinations, **7,194 had above-average demand**. These were then evaluated based on demand variability, with **3.84 units** representing the average standard deviation within the high-demand group. This resulted in **2,116 combinations** that had both above-average demand and above-average variability.

The final analysis highlighted the store-product combinations with the highest demand variability within this group. One particularly notable pattern was **FOODS_3_090**, which appeared repeatedly across multiple stores. For example, its average daily demand was approximately **130.95 units in CA_3**, accompanied by a standard deviation of **108.61 units**. Other stores also showed substantial variability for the same product, indicating that the demand pattern is not isolated to a single store.

This finding makes the analysis useful from an operational planning perspective because it distinguishes **high-volume demand from high-volume but less predictable demand**. A product that sells consistently at a high level may be relatively straightforward to plan for, whereas a product with both substantial demand and significant fluctuations may require closer monitoring.

Business Implication

High-demand and high-variability products represent an important **planning-priority group**. Unexpected changes in their demand can have a greater effect on overall unit requirements because these products already operate at relatively high demand levels.

The analysis does **not** establish that these products experienced stockouts or inventory shortages, since actual inventory and stockout information was not available. Instead, it identifies combinations where demand variability may make forecasting and replenishment more challenging.

Recommendation

Walmart could give **additional forecasting and replenishment attention to high-demand products with substantial demand variability**. These products can be monitored more frequently and evaluated using historical demand patterns at the individual store-product level rather than relying solely on broad product-level averages.

2.4 Most Intermittent Products

The analysis also identified a group of products with **highly intermittent sales activity**, meaning that these products recorded sales on only a relatively small proportion of the days covered by the dataset. Rather than measuring intermittency simply by the number of units sold, the analysis used the **percentage of days on which a product recorded sales** across the 1,913-day period. This provides a more meaningful way to compare products with different demand volumes.

The results show substantial differences in sales frequency. For example, **FOODS_3_296 recorded sales on only 192 of the 1,913 days**, equivalent to **10.04% of the analysis period**. This means that the product had no recorded sales on approximately 90% of the days. Other products in the analysis also showed low selling-day percentages, demonstrating that demand is not equally continuous across the product portfolio.

This finding adds an important dimension to the earlier demand-concentration and variability analysis. **A product can have low sales frequency even when it remains relevant within the product portfolio**. Therefore, evaluating products only by total units sold may overlook differences in how frequently demand occurs.

Business Implication

Intermittent products may require a different planning perspective from products that generate sales consistently. Their low frequency of observed sales makes simple assumptions based on continuous daily demand less appropriate.

However, intermittent demand **does not automatically mean that a product should be discontinued or stocked at a lower level**. The analysis identifies an unusual sales pattern; it does not provide enough information to determine whether the pattern is caused by customer demand, product availability, product lifecycle, promotions, or other factors.

Recommendation

Walmart could use **sales frequency as an additional product-segmentation factor** when reviewing demand and replenishment strategies. Products with highly intermittent sales can be flagged for further evaluation rather than being treated in the same way as consistently selling products.

Combining **sales frequency, demand volume, and demand variability** would provide a more complete basis for determining how different products should be monitored and planned.

Key takeaway: The analysis demonstrates that demand differs not only in *how much* a product sells, but also in *how consistently* it sells—providing another useful dimension for demand planning.

2.5 Store Performance and Growth

The store-level analysis shows that demand and year-over-year performance are **not uniform across the store network**. The analysis therefore provides a useful basis for understanding where demand is concentrated and where store performance has changed over time.

The annual growth results show meaningful differences between stores. For example, **WI_2 recorded particularly strong growth in the earlier years**, increasing by approximately **79.91% in 2012** and **37.57% in 2013**. In contrast, **WI_3 experienced negative growth in both 2013 (-16.44%) and 2014 (-12.31%)**, before returning to positive growth in 2015. These differences demonstrate that stores can follow substantially different demand trajectories even within the same overall business.

The analysis also shows that store performance should be interpreted over time rather than from a single year's result. Some stores experienced periods of strong growth followed by slower growth or decline, while others showed more stable patterns. This provides a useful signal for identifying stores that may warrant closer examination of their demand patterns.

Business Implication

The differences between stores suggest that **a completely uniform demand-planning approach may not reflect actual store-level demand patterns**. Stores with different demand levels or growth trajectories may require different forecasting assumptions and planning attention.

However, the **2016 results require caution**. The dataset ends on **April 24, 2016**, meaning that 2016 represents only a partial year. The large negative year-over-year values observed for 2016 therefore **should not be interpreted as full-year declines in store performance**.

Recommendation

Walmart could incorporate **store-level demand and historical growth patterns** into its planning process. Stores with consistently strong demand or notable changes in performance can be prioritized for closer monitoring and further investigation.

For future analysis, store-level demand could also be combined with additional information—such as local conditions, promotions, or actual inventory availability—to better understand the reasons behind differences in store performance.

Key takeaway: The analysis demonstrates that store performance is not homogeneous, creating an opportunity for **more localized and data-informed demand planning** rather than relying entirely on network-wide assumptions.

2.6 Limited Difference Between Holiday and Regular-Day Demand

The holiday and event analysis found **no substantial difference in overall unit demand between regular days and days associated with recorded holidays or events**. This suggests that, at the overall dataset level, holidays and events did not produce a strong enough change in demand to be considered a major driver of total demand.

This is a **useful analytical finding**. The analysis tested whether calendar events represented an important source of demand variation, and the results indicate that they were **not a dominant factor in overall demand** within this dataset.

Business Implication

The relatively small difference means that Walmart should **not assume that every holiday or event requires a major adjustment to overall demand expectations**. Broader product-, store-

, and demand-level patterns appear to provide more meaningful signals for planning than the simple distinction between holiday and regular days.

This does not mean that individual products or specific events cannot experience meaningful changes in demand. It only means that the **overall effect was limited in the analysis performed.**

Recommendation

Holiday and event information can still be retained as a **secondary planning factor**, particularly when planning specific products or events with historically unusual demand. However, it should not be given the same priority as stronger signals identified in this analysis, such as **product demand concentration, category contribution, and high-demand demand variability.**

Key takeaway: The analysis was productive even though the difference was small—it helped establish that **holidays/events were not a major explanation for overall demand variation in this dataset.**

2.7 Category Demand Changes Over Time

The category trend analysis showed how the demand mix changed across **FOODS, HOUSEHOLD, and HOBBIES** from 2011 to 2016. **FOODS remained the largest contributor to demand throughout the period**, while HOUSEHOLD and HOBBIES contributed smaller shares.

The analysis also showed that the category shares **changed over time rather than remaining exactly constant.** This means the overall demand mix was not completely stable, even though FOODS continued to dominate.

Business Implication

FOODS should remain the main category for demand planning, while changes in the shares of HOUSEHOLD and HOBBIES should be monitored to identify shifts in customer demand.

Recommendation

Track **category demand share and total units over time** to identify changes in the product mix and adjust planning priorities when meaningful shifts occur.

Key takeaway: The analysis confirmed that **FOODS consistently drives the largest share of demand**, while the relative contribution of the other categories changes over time.

3. Strategic Business Recommendations

The analysis indicates that demand planning can be made more effective by giving different levels of attention to products and stores based on their demand contribution, variability, and sales frequency. The following recommendations focus on the areas where the analysis provides the strongest evidence.

3.1 Prioritize High-Contribution Products

The top 10% of products account for **51.14% of total unit demand**, showing that a relatively small part of the product portfolio has a major influence on overall demand.

Recommendation:

Give these high-contribution products greater forecasting, replenishment, and monitoring attention. Prioritizing them allows planning resources to focus on products where demand decisions can have the greatest overall impact.

3.2 Closely Monitor High-Demand Products with High Variability

The analysis identified **2,116 store-product combinations** with both above-average demand and above-average demand variability. These combinations represent products that generate meaningful demand but also show substantial fluctuations.

Recommendation:

Flag these combinations for closer demand monitoring and forecasting. Planning should consider **store-product level demand patterns** rather than relying only on overall product averages.

This is a **planning-risk signal**, not evidence that these products experienced stockouts, because actual inventory data was not available.

3.3 Use Different Planning Approaches for Intermittent Products

Some products sell on a very small proportion of days. For example, **FOODS_3_296 recorded sales on only 10.04% of the 1,913 days** in the dataset.

Recommendation:

Identify highly intermittent products as a separate planning group and avoid applying the same assumptions used for consistently selling products. Their low sales frequency should be considered when forecasting and reviewing replenishment requirements.

The analysis does **not** indicate that these products should automatically receive less inventory; further information would be required to make that decision.

3.4 Give Greater Planning Attention to FOODS

FOODS represents **68.63% of total unit demand**, making it the largest demand category by a significant margin.

Recommendation:

Give greater category-level monitoring and forecasting attention to FOODS, particularly where its products also appear among the high-demand or high-variability groups. This combines the category-level finding with the more detailed product-level analysis.

4. Priority Action Plan

The analysis suggests that not all areas require the same level of attention. The following priorities focus on where the findings indicate the greatest potential value.

Priority 1 — Focus on High-Contribution Products

Why: The top 10% of products generate **51.14% of total demand**.

Action: Prioritize these products in forecasting, replenishment reviews, and demand monitoring.

Priority 2 — Monitor High-Demand, High-Variability Combinations

Why: **2,116 store-product combinations** showed both above-average demand and above-average variability.

Action: Give these combinations closer monitoring and use store-product level demand patterns when planning.

Priority 3 — Segment Intermittent Products

Why: Some products sell on only a small percentage of days, with **FOODS_3_296 selling on just 10.04% of days**.

Action: Treat highly intermittent products as a separate planning group and evaluate them using demand-frequency information.

Priority 4 — Maintain Strong Focus on FOODS

Why: FOODS contributes **68.63% of total unit demand**.

Action: Give the category greater attention in demand monitoring, particularly for products that are also high-demand or highly variable.

Overall Priority

The main message is simple: **focus planning effort where the data shows the greatest demand concentration and uncertainty**. This means prioritizing important products first, then considering how stable and frequent their demand is, while giving appropriate attention to the stores and categories where demand is concentrated.

5. Limitations & Responsible Interpretation

The findings in this analysis provide useful demand-planning insights, but they should be interpreted within the limits of the available data.

- **Partial 2016 data:** The dataset ends on April 24, 2016. Therefore, 2016 should not be treated as a complete year when comparing annual performance.
- **Units sold, not financial performance:** The analysis focuses on unit demand. Revenue, profit, and margins were not evaluated.
- **No actual inventory or stockout data:** The analysis identifies demand patterns and potential planning challenges, but it cannot confirm whether products experienced stockouts or excess inventory.
- **Limited explanation of demand drivers:** The data does not provide enough information to determine why particular products or stores experienced changes in demand.
- **Holiday/event effects:** The overall difference between regular and holiday/event days was limited, so these factors should not be treated as major drivers of total demand based on this analysis alone.

These limitations do not reduce the value of the analysis; rather, they define **how the findings should be used**. The results are best viewed as data-driven signals that can support further investigation and planning decisions.

6. Conclusion

The analysis provided several clear insights into Walmart's demand patterns. Demand is highly concentrated, with the **top 10% of products contributing 51.14% of total demand**, while **FOODS contributes 68.63%** of overall unit demand. The analysis also identified high-demand products with substantial variability and products with highly intermittent sales activity, showing that demand characteristics differ significantly across the product portfolio.

These findings provide a practical basis for **prioritizing demand-planning efforts**. High-contribution products should receive greater attention, high-demand but highly variable products should be monitored closely, and intermittent products should be considered separately when planning demand and replenishment.

Overall, the project demonstrates how sales data can be transformed into **clear business signals rather than just descriptive statistics**. While additional information such as inventory levels, stockouts, revenue, and profit would be needed for deeper operational and financial decisions, the analysis provides a strong foundation for more focused and data-informed planning.